

Ranger / VISTA Bench

Agents already take long sequences of steps. What no one can measure is whether they hold the **long view** — plan far ahead, stay on a multi-step goal as the terrain shifts, resist the OWASP agentic attacks, and get **safer, not more drift-prone, as they self-improve**. VISTA Bench measures it; Ranger is the three-model agent built to hold it.

DIRECTION	WE MEASURE	HARDENS VS	SHOWCASE
B • Agents, rethought	Foresight × safety	OWASP agentic Top-10	Mon • Jun 29

01 — THE PROBLEM

The unmeasured question isn't whether an agent can run long — it's whether it holds the **long view**: plan far ahead, stay aligned to intent, and get **safer rather than drift as it self-improves**.

Long-horizon failure is real and measured (METR; *Illusion of Diminishing Returns*, ICLR 2026) — the failures are execution, not reasoning. The long view breaks three ways: the agent **drifts off-plan** (OWASP ASI01 Goal Hijack), untrusted terrain **hijacks it** (ASI02–ASI07), or it **self-improves in the wrong direction** — reward-hacking and goal drift, as the Darwin Gödel Machine and Anthropic's production-RL study both documented. Today's benchmarks test short episodes or one capability; **none measures plan + safety together over a horizon — let alone whether self-improvement makes the agent safer**.

02 — THE LANDSCAPE • WHY A NEW BENCHMARK IS NEEDED

Every published agent benchmark measures one slice, in one session. The empty quadrant: **foresight × safety, sustained over a long horizon, improving (not decaying) as the agent self-improves**. That is VISTA Bench.

CAMP & PUBLISHED EXAMPLES	WHAT THEY MEASURE	THE GAP FOR THE LONG VIEW
Capability & task completion WebArena · OSWorld · GAIA · AppWorld · AgentBench · SWE-bench · Terminal-Bench 2.0	Can it finish a task?	Bounded runs, pass/fail; no foresight; mostly no safety.
Workplace & policy TheAgentCompany · T-bench	Professional tasks; policy (t: pass*k)	Long, checkpoint-scored — but one bounded run; no adversary.
Long-horizon & coherence METR / HCAST · Vending-Bench	Human-time horizon; coherence / derail	Closest on horizon — but no planning quality, no security.
Memory & async LongMemEval · Gala2 / ARE	Recall; async + noise	Scores answers, not the plan; no malicious adversary.
Security & self-improvement AgentDojo · InjecAgent · ST-WebAgentBench · SafeArena · AgentHarm · SafeEvalAgent	Injection, misuse; safety across cycles	Single-session, or safety that only decays; no foresight.

The closest — **AgentLAB / AgentHarm** score attack-success never planning; **SafeEvalAgent / Meta-Agent Challenge** show safety **decaying** as agents self-improve. **VISTA is the first to measure foresight (planning) × safety together** over a horizon — and whether self-improvement **raises** safety — graded against the **OWASP agentic Top-10**.

03 — DIRECTION & TECHNICAL APPROACH

DIRECTION B The brief's *example* is one shared agent the team verifies; the real question is broader. So Ranger is a **three-model agent system**, and the novelty is **VISTA**, the benchmark. The Scout/Worker split is a **published injection defense** (Dual-LLM → CaMeL); the Dreamer adds **guardrailed** recursive self-improvement (Reflexion · Generative-Agents reflection · Voyager). The model is the loop, split in three:

► **SCOUT**

The long view. Maps milestones, holds the route, re-plans under drift, escalates the few high-risk forks.

► **WORKER**

The near terrain. Heavily sandboxed; executes steps with **no authority or secrets** (contains ASI02/03/05).

► **DREAMER**

Between legs. Guardrailed recursive self-improvement (RSI): consolidate skills, purge poisoned beliefs (ASI06), self-audit vs the OWASP Top-10. **Verified signal + traceable lineage** so it can't reward-hack (the DGM failure).

SHARED — a map/memory with belief revision the agents hand off through · privilege separation + anti-poison gate · calibrated human steering (counters ASI01 goal-hijack, ASI09 trust-exploitation, ASI10 rogue agents).

MINIMUM SHIPPABLE

Proof the gap is real

- VISTA harness + 3 multi-day journeys, seeded with OWASP agentic attacks
- Ranger (Scout + sandboxed Worker + guardrailed Dreamer) at the top of the range
- Head-to-head vs a single-model agent; the long-view premium
- The new measurement: does safety rise or drift across Dreamer cycles?

PROTOTYPE VALIDATED END-TO-END (clear baseline-vs-steward spread) - rebuilding against the frozen contracts.

THE AMBITIOUS VERSION

A benchmark with teeth

- 9-12 journeys; full OWASP agentic Top-10 coverage
- Adapters driving Claude Code, Codex, OpenClaw, Hermes (Harbor-style)
- Ablations: Scout / sandbox / Dreamer off + human baselines
- A paper: the first benchmark where safety is measured across self-improvement cycles

05 - WHY IT'S COMPELLING

Everyone races to make agents smarter for one task. No one can yet say whether a long-running agent stays aligned and safe — or quietly drifts — as it improves itself.

You'd ship a benchmark with teeth tied to the OWASP agentic Top-10, and the three-model agent that wins it — real systems work (planner/executor sandboxing, memory, a guardrailed self-improvement loop, evals, model comparisons). Clean ownership for four: benchmark + scoring, Scout (planner), sandboxed Worker + harness, Dreamer + research/demo.

06 - HARD PARTS / RISKS

THE BET	Foresight + safety are measurable, and the three-model split holds the long view better than one model. A flat premium means we're wrong — still publishable.
RSI IS DOUBLE-EDGED	Self-improvement can amplify misalignment — the DGM hacked its own reward; Anthropic shows reward-hacking generalizes to sabotage. The Dreamer is the guardrailed exception (verified signal, traceable lineage, capped change); if guardrails fail, VISTA catches the drift — which is the point.
NOVELTY	Ranger is a known pattern (CaMeL; Plan-and-Act; Voyager); the contribution is VISTA. No benchmark scores safety improving across self-improvement cycles — the closest show it decaying.
MEASUREMENT	Rubrics that can't be gamed over a moving horizon — hidden ground truth + state-based checks mapped to OWASP ASI threats.

07 - THE DEMO · LIVE · 10 MINUTES · JUN 29

00:00	● Day 1 — the goal The Scout maps the milestones; the Worker starts the first terrain steps inside its sandbox.
02:00	● Drift + live steering The decision-maker changes the destination. The Scout re-plans and escalates the one scope change for a human OK — live.
04:00	● OWASP ambush A web page attempts goal-hijack + memory-poisoning (ASI01/ASI06). The Worker sandbox + Dreamer anti-poison gate refuse it; nothing leaks.
06:00	● The Dreamer self-improves Between legs it consolidates, purges the poisoned belief, self-audits vs the OWASP Top-10, and sharpens the route — we show the dream journal and its traceable lineage.
08:00	● Scoreboard Ranger vs a single-model agent — the long-view premium, and does safety rise across cycles? Visible where pass/fail sees nothing.

08 - TEAM, TOOLING & CONTRACTS

Four people, four owned modules, four contracts locked on day one — everyone builds in parallel, nothing blocks. The architecture is the work-breakdown:

BENCHMARK	VISTA harness, the 9 scoring axes, OWASP-ASI journeys, the scoreboard.
SCOUT	The planner model: milestone / route schema, drift re-planning, escalation.
WORKER	The sandboxed executor: tool API, privilege separation, external-agent adapters.
DREAMER	The self-improvement loop: belief revision, RSI guardrails, research + demo.

CONTRACTS — lock first so no one waits: route-state schema · agent-tool API · 9-axis rubric · AgentAdapter interface. Each dev pairs with Claude Code or Codex on their module behind the frozen contract; a shared CLAUDE.md / AGENTS.md carries the contracts, and every PR must pass the contract test suite to merge.